

SIGNAL & DATA ANALYSIS IN NEUROSCIENCE 2020 PROBABILITY & STATISTICS REHEARSAL

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Outline

- Administrations
- Probability & Random variables (RV): definitions & basic terms.
- Common distributions.
- Statistics: inference based upon RV analysis.

Administrations

- Course website: <https://www.ibglab.org/sda-2020>
Password: sda2020
- Course Email: biu.sigproc@gmail.com
- Tigbur: Questions sent by Friday 1PM will be attended.
No emails, no tigbur..
- Exercises
 - Submission time: Thursday 8AM.
 - Make sure that all figures have axis labels, units, legends, titles.

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Probability & Random variables (RV): definitions & basic terms.

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Probability definition

- Probability theory deals with the study of random phenomena, which under repeated experiments yield different outcomes that have certain underlying patterns about them.
- Probability is marked by $P(a)$ when,
 - 1) The sample space (S) is the collection of all possible events and is marked by a capital letter.
 - 2) $P(A)$ is the probability for event 'A' to occur. Sometimes events are marked by lower case letters.
- Venn diagrams: demonstrate $P(A \cup B)$, $P(A \cap B)$
- The axioms of probability:
 - (i) $P(A) \geq 0$ (Probability is a nonnegative number)
 - (ii) $P(\Omega) = 1$ (Probability of the whole set is unity)
 - (iii) If $A \cap B = \phi$, then $P(A \cup B) = P(A) + P(B)$.

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Random variables (RV) definitions

- **Random variable (RV)** is a function that maps events or outcomes from the **sample space** to real numbers.
- Usually - the random variable is marked by a capital letter (e.g. X) with its values marked in lower case (e.g. x)

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Example: probability & RV

- $s \in \{\text{set of outcomes of ten coin tosses}\}$

Probability for 's': $P(s) = \frac{1}{2^{10}}$

RV: $X(s)$ is the number of heads in the sequence

$s = \text{HHTTTHTTTT} \rightarrow X(s) = 4$

$$P(X = x) = \frac{\binom{10}{x}}{2^{10}} \quad \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- $s \in \{\text{mean firing rate of a neuron over a time period}\}$

$$X(s) = \begin{cases} 0 & \text{if firing frequency} < 5 \text{ spikes/sec,} \\ 1 & \text{else .} \end{cases}$$

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Cumulative Distribution Function (CDF)

- Let us denote the CDF for a RV X as $F_X(x)$

$$F_X(x) = P(X \leq x), \quad F(x) = \int_{-\infty}^x f(t) dt.$$

- Properties of the CDF:

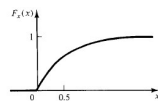
$$F_X(x = -\infty) = 0$$

$$F_X(x = \infty) = 1$$

$$\text{if } x_1 < x_2 \Rightarrow F_X(x_1) \leq F_X(x_2)$$

$$P(x > x_1) = 1 - F_X(x_1)$$

$$P(x_1 < x < x_2) = F_X(x_2) - F_X(x_1)$$



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Probability density function (pdf)

- The probability density function **pdf** is defined as

$$f_X(x) = \frac{dF_X(x)}{dx}$$

- Properties of pdf:

$$f_X(x) \geq 0$$

$$F_X(x_1) - F_X(x_2) = \int_{x_1}^{x_2} f_X(x) \cdot dx$$

$$\int_{-\infty}^{\infty} f_X(x) \cdot dx = 1$$

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Discrete RV

- X is a discrete random variable (or actually has a discrete distribution) if it has a finite set of different values $x_1, x_2, \dots, x_k$ or at most an infinite sequence $x_1, x_2, \dots$

- In this case the probability density function is defined as:

$$F_X(x_0) = P(X \leq x_0)$$

$$f_X(x) = p(x)$$

$$\sum_{i=1}^{\infty} f_X(x_i) = 1$$

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Common distributions

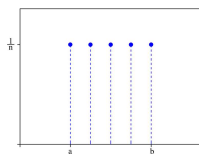
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Discrete uniform distribution

- In a uniform distribution X has an equal probability to assume each of the k values $x_1, x_2, \dots, x_k$

$$f(x) = \begin{cases} \frac{1}{k} & x \in \{x_1, x_2, \dots, x_k\} \\ 0 & \text{otherwise} \end{cases}$$

- **Example:**
 $n = b - a + 1$



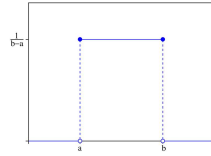
Example:
Draw the pdf & CDF for a discrete RV

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Continuous uniform distribution

- Uniform distribution: X is equally likely to occur along the segment [a,b]

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$



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Discrete Binomial distribution

- Binomial distribution: If the sample space includes n samples of two independent options with probability p & (1-p).

X is the number of occurrences of the first options

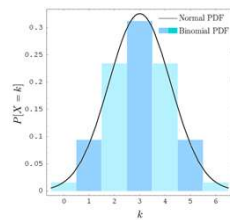
$$f(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & \text{for } x = 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

- Example:

n=6 p=0.5

Six tosses of a coin

X is the number of heads



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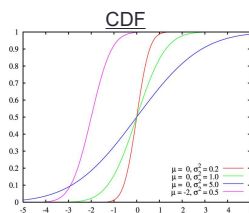
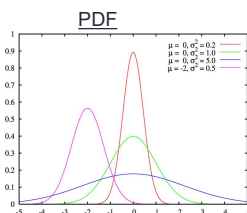
Normal distribution

- A random variable is termed Gaussian if:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$F(x) = \frac{1}{2} \left(1 - \operatorname{erf} \frac{x-\mu}{\sqrt{2}\sigma} \right)$$

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-z^2} dz$$



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The central limit theorem

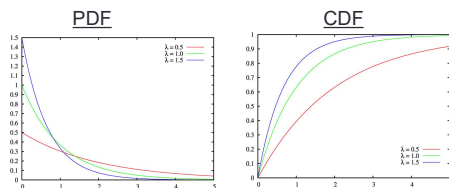
- **Central Limit Theorems** are a set of converging results in probability theory.
- The most important and famous result is called simply **The Central Limit Theorem** which states that if the **sum of independent identically distributed random variables** has a finite variance, then it will be **approximately normally distributed**.
- Generalizations exist for some cases of none independent and non identical random variables.
- Since many real processes yield distributions with finite variance, this explains the ubiquity of the normal distribution.

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Exponential distribution

- A random variable is exponential if:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases} \quad (\lambda > 0) \quad F(x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases} \quad (\lambda > 0)$$



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Additional items

- Joint distributions.
- Conditional distributions.
- Multivariate distributions.

Reading material: Probability and Statistics / M.H. DeGroot, MJ Schervish Chapters 2.1-2.2 & 3.4-3.7

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Statistics: inference based upon RV analysis

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Motivation for operations on RV

- The distribution of X contains **all** the probabilistic information about X . However it is usually too cumbersome for presenting it.
- Different **operations** on X may provide a good idea about X without describing the full distribution.

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Expectation (AKA $\mu, \langle X \rangle, \bar{X}$)

- The expectation of a discrete RV: $E[X] = \sum_x x \cdot f_X(x)$
- The expectation of a continuous RV: $E[X] = \int_{x=-\infty}^{\infty} x \cdot f_X(x)$
- Properties:
 1. $Y = aX + b \Rightarrow E(Y) = aE(X) + b$
 2. $P(X \geq a) = 1 \Rightarrow E(X) \geq a$
 $P(X \leq b) = 1 \Rightarrow E(X) \leq b$
 3. $E(X_1 + \dots + X_n) = E(X_1) + \dots + E(X_n)$
 4. $E\left(\prod_{i=1}^n X_i\right) = \prod_{i=1}^n E(X_i)$
(if $X_1, \dots, X_n$ independent)
 5. $E(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)dx$

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Example: expectation

- The pdf of the random variable X

$$f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

- The expectation is

$$E(X) = \int_0^1 x \cdot 2x dx = \int_0^1 2x^2 dx = \frac{2}{3} x^3 \Big|_0^1 = \frac{2}{3}$$

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Moments

For each random variable X and positive integer n:

- The nth order **moment** of X is defined as:

$$m_n = \overline{X^n} = E(X^n), \quad n \geq 1$$

- The nth order **central moment** of X is defined as:

$$\mu_n = E[(X - \mu)^n]$$

when $\mu = m_1$.

- The 1st moment is called the mean
- The 2nd central moment is called the variance

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Variance

- For a RV X with mean $\mu = E(X)$ the variance of X is:

$$\text{var}(X) = \sigma^2 = E[(X - \mu)^2]$$

- The standard deviation is defined as the positive square root of the variance: $\text{std}(X) = \sqrt{\text{var}(X)} = \sigma$

- Properties:

$$1. \text{var}(X) = 0 \Leftrightarrow P(X = c) = 1$$

$$2. \text{var}(aX + b) = a^2 \text{var}(X)$$

$$3. \text{var}(X) = E(X^2) - [E(X)]^2$$

$$4. \text{var}(X_1 + \dots + X_n) = \text{var}(X_1) + \dots + \text{var}(X_n) \\ (\text{if } X_1, \dots, X_n \text{ independent})$$

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Example: variance

- The pdf of the random variable X

$$f(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

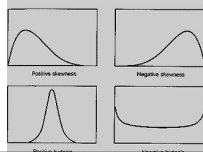
- The variance is

$$\text{Var}(X) = \int_0^1 (x - \frac{2}{3})^2 \cdot 2x dx = \int_0^1 2x^3 - \frac{8}{3}x^2 + \frac{8}{9}x dx = \left[\frac{1}{2}x^4 - \frac{8}{9}x^3 + \frac{4}{9}x^2 \right]_0^1 = \frac{1}{18}$$

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Higher moments

- Higher moments are less commonly used but will still play a role in the course.
- The 3rd standardized moment is called the **skewness** and reflects the asymmetry of the distribution around the mean
- The 4th standardized moment is called the **kurtosis** and reflects to "peakedness" of the distribution
- The skewness of the normal distribution is 0 & the kurtosis is 3.



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Covariance: Relation between RV

- Covariance** –the measure of how much two variables vary together. X,Y are RV.
 $E(X)=\mu_X$, $E(Y)=\mu_Y$, $\text{Var}(X)=\sigma_X^2$, $\text{Var}(Y)=\sigma_Y^2$
 $\text{Cov}(X,Y)=E[(X-\mu_X)(Y-\mu_Y)]$
- The covariance is
 - more positive for each pair of values which differ from their mean in the same direction
 - more negative with each pair of values which differ from their mean in opposite directions.
- Properties:
 - $\text{Cov}(X,X)=\text{Var}(X)$, **$\text{Cov}(X,Y)=\text{Cov}(Y,X)$** , $\text{Cov}(cX,Y)=c\text{Cov}(X,Y)$
 - $\text{Cov}(X,Y)=E(XY)-\mu_X\mu_Y$
 - If X & Y are independent $\text{Cov}(X,Y)=\rho(X,Y)=0$
 - $\text{Var}(X+Y)=\text{Var}(X)+\text{Var}(Y)+2*\text{Cov}(X,Y)$

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Covariance: multi dimensional vectors

- We will use the following notation which is compliant with MATLAB. **For matrices:**
 - each row is an observation
 - each column is a variable
- Example: X represents a RV with a single variable, hence $\text{size}(X) = (N,1)$
- Example: X represents a row vector of RV with N samples each with M variables, hence $X_{[N \times M]}$
- $\text{Cov}(X_{[N \times M]}, Y_{[N \times M]}) = E[(X - \mu_X)'(Y - \mu_Y)]$
- Note: the used notation is compliant with Matlab $\text{cov}(x,y)$ notation.

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Correlation

- Correlation is a measure of association which is not driven by the magnitude of X & Y

$$\rho(X,Y) = \frac{\text{Cov}(X,Y)}{\sigma_x \sigma_y}$$

- Cauchy Schwartz inequality: $|E(XY)|^2 \leq E(X^2)E(Y^2)$.
if $U = X - \mu_X$ and $V = Y - \mu_Y \rightarrow [\text{Cov}(U,V)]^2 \leq \sigma_U^2 \sigma_V^2$
Thus $-1 \leq \rho(X,Y) \leq 1$

- Properties:
 - $-1 \leq \rho(X,Y) \leq 1$

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Correlation & linearity

- Linear dependent random variables
Example: If X is a random variable and $Y = aX + b \rightarrow \rho(X,Y) = 1$ for $a > 0$ and $\rho(X,Y) = -1$ for $a < 0$
- **Correlation only measures linear relationship.**
- Non-linear dependent random variables
Example: If X is a random variable taking the values -1,0,1 with equal probabilities and $Y = X^2$
 $E(XY) = E(X^3) = 0$ and $E(X) = 0 \rightarrow E(X)E(Y) = 0$
Thus $\text{Cov}(X,Y) = 0 \rightarrow \rho(X,Y) = 0$
